

# Timothy J Scholtz

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## Education

**Ph.D. Computer Engineering**, Virginia Tech 2025 - Present

**Research Area:** Hardware-Software Co-Design and Analysis

**Bachelor of Science, Computer Engineering**, University at Buffalo GPA: 3.88 2021- 2025

## Research Experiences

**Location:** Virginia Tech – FoRTE Lab

**Date:** 2025 June – Current

**Title:** Research Assistant

**Supervisor:** Prof. Matthew Hicks

- Researching hardware-software interactions and exploring the design and problem space.
- Bespoke processor generation leveraging symbolic execution. Built tool to identify and remove redundant gates in processors, achieving average 55% reduction in power and area.
- Edge coverage limit estimation using symbolic execution and software analysis for Fuzzing applications.

**Location:** Center for Computational Research

**Date:** 2024 May – August

**Title:** Research Assistant

**Supervisor:** Dr. Joseph White

**Research Project:** Augmenting a developmental virtual assistant to suit our team's purpose:

- Built a pipeline to generate, clean, and embed data in order to build a RAG database to supplement the support team's AI chatbot with knowledge from our team's work with an above 90% accurate answer rate.
- Added features to XDMoD web portal, such as code generation to replicate charts in python environment.

**Location:** School of Engineering – University at Buffalo

**Date:** 2023 April – August

**Title:** Research Assistant

**Principal Investigator:** Prof. Wenyao Xu

**Research Project:** Developing an automated speech recording and analysis tool:

- Integrated multiple speech-to-text tools with a custom sentence-combining algorithm; improved accuracy through device testing and analysis, reduced Word Error Rate by 5% via comparative evaluation.
- Led weekly team meetings and presented project outcomes to the department.

**Location:** School of Computer Science – University at Buffalo

**Date:** 2021 December – 2022 August

**Title:** Research Assistant

**Principal Investigator:** Dr. Weihang Wang

**Research Project:** Automated Detection of Compilation Differences Between Compilers

- Developed a differential software testing framework to analyze compilation discrepancies between 2 compilers, finding inconsistencies and failures in 7% of the input software. Inspected and categorized failures.

## Paper & Research Outputs

- **Timothy Scholtz**, Matthew Hicks. "Leveraging Software Analysis for Better Bespoke Processor Generation," *ASPLOS 2027* (Under Revision).
- **Timothy Scholtz**, Matthew Hicks. "N-State Runtime Recovery for Radiation-Induced Soft Errors on Commodity Hardware," Cisco Semiconductor Reliability Research Gift, 2026 (Under Review).
- **Timothy Scholtz**, Matthew Hicks. "Knowing When to Stop: Coverage Upper Bounds for Fuzzing via Symbolic Execution," Air Force Research Laboratory (AFRL), 2026 (Under Review).

## Work Experiences

- **Teaching Assistant:** 2023 September – 2026 May

Worked as a TA for Intro to CS, Embedded Systems, Digital Systems, and Computer Organization.

- **Tutor:** 2022 September – 2023 May

Helped student with introductory Math, Physics and CS.

- **Gravatronics:** 2021 January – April

Interned at a BMS startup. Prototyped circuits, coded scripts, managed inventory, and wrote documentation.

## Skills

**Programming Languages:** Python, C, Verilog, Bash, Scala, Java, SQL, JavaScript, Assembly

**Tools:** LaTeX, Arduino, Audio Processing, MySQL, Linux, Microsoft Office tools, Git, Visual Studio, Logisim

**Soft Skills:** Problem Solving, Analytical, Dependability, Communication, Organization, Debugging, Self-starter